1. Clinical Question & Core Hypothesis

Clinical Question

The graph highlights YAP as a mechanoresponsive effector regulated by branching morphogenesis and kinase signaling, but it is unclear when/mechanistically how stromal or morphogenetic tissue context converts YAP activation into therapy-resistant disease. In other words: why do some stroma-rich tumors fail to respond to therapy despite a non-neoplastic microenvironment? 2) Tier 1 paths used: (a) Branching morphogenesis → regulates → YAP (mechanotransduction link); (b) YAP → mutated_in / dysregulated_in → cancer (causal role in progression). I built on these because they form the strongest mechanistic backbone tying tissue architecture/mechanics to oncogenic YAP programs. 3) Tier 2 evidence supporting the claim: kinase signaling and kinase inhibition (staurosporine) appear as modulators of YAP activity, suggesting that upstream kinases (Hippo-pathway kinases) can toggle the YAP-driven state; vascular/branching biology provides plausible sources of mechanical cues. 4) Why I chose therapy_response as the endpoint: the provided KG emphasizes signaling/state changes (YAP activation) that commonly drive resistance phenotypes; therapy_response (endocrine therapy resistance) is the most conceptually appropriate clinical consequence to test a YAP-mediated reprogramming hypothesis within available breast-cancer datasets. 5) Deliberate exclusions: I treated the skin-permeability, mpMRI, ALP/creatinine, and radiation/salivary-gland branches as cross-domain noise for this question and excluded them from the core hypothesis to maintain mechanistic focus on tissue mechanics → YAP → therapy resistance. I also excluded any specific kinase gene names because none were explicitly provided in the input?

Working Hypothesis

High stromal-to-tumor ratio (TSR) mechanically activates YAP signaling in breast tumor cells, leading to a YAP-driven transcriptional program that dampens estrogen-receptor–mediated signaling and promotes cell survival under endocrine therapy. Consequently, tumors with both high TSR and elevated YAP activity will exhibit reduced responsiveness to endocrine therapy (endocrine therapy resistance) compared with tumors with low TSR or low YAP activity. This relationship is at least partly mediated by upstream Hippo-pathway kinases; pharmacologic or genetic inhibition of YAP will restore or improve endocrine therapy sensitivity in high-TSR contexts. The hypothesis will be tested by (i) demonstrating an interaction between TSR and YAP activity predicting endocrine therapy response in breast cancer cohorts, (ii) showing that mechanical cues (ECM stiffness) drive YAP activation and endocrine resistance in ER-positive breast cancer cell models, and (iii) proving that YAP inhibition reverses resistance in vitro and in vivo models with high stromal content.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	YAP
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

YAP: YAP activation) that commonly drive resistance phenotypes; therapy_response (endocrine therapy resistance) is the most conceptually appropriate clinical consequence to test a YAP-mediated reprogramming hypothesis within available breast-cancer datasets.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	Branching morphogenesis (mechanical cue)	Supporting modifier from knowledge graph
Secondary	Kinase signaling (Hippo-pathway kinases)	Supporting modifier from knowledge graph
Secondary	Staurosporine (broad kinase inhibitor probe)	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Branching morphogenesis (mechanical cue): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- Kinase signaling (Hippo-pathway kinases): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.